

NAME

apdefault -- Set default aperture parameters for the package

USAGE

apdefault

PARAMETERS

lower = -5., upper= 5.

Default lower and upper aperture limits relative to the aperture center. These limits are used for apertures found with **apfind** and when defining the first aperture in **apedit**.

apidtable = ""

Aperture identification table. This may be either a image. A text file consisting of lines with an aperture number, beam number, and aperture title or identification. An image will contain the keywords SLFIBnnn with string value consisting of aperture number, beam number, optional right ascension and declination, and aperture title. This information is used to assign aperture information automatically in **apfind** and **apedit**.

Default Background Subtraction Parameters

b_function = "chebyshev"

Default background fitting function. The fitting function types are "chebyshev" polynomial, "legendre" polynomial, "spline1" linear spline, and "spline3" cubic spline.

b_order= 1

Default background function order. The order refers to the number of terms in the polynomial functions or the number of spline pieces in the spline functions.

b_sample = "-10:-6,6:10"

Default background sample. The sample is given by a ranges each separated by either whitespace or commas. The string "*" refers to all points. Note that the background coordinates are relative to the aperture center and not image pixel coordinates so the endpoints need not be integer.

b_naverage= -3

Default number of points to average or median. Positive numbers average that number of sequential points to form a fitting point. Negative numbers median that number, in absolute value, of sequential points. A value of 1 does no averaging and each data point is used in the fit.

b_niterate= 0

Default number of rejection iterations. If greater than zero the fit is used to detect deviant fitting points and reject them before repeating the fit. The number of iterations of this process is given by this parameter.

b_low_reject= 3., b_high_reject= 3.

Default background lower and upper rejection sigmas. If greater than zero points deviating from the fit below and above the fit by more than this number of times the sigma of the residuals are rejected before refitting.

b_grow = 0.

Default reject growing radius. Points within a parameter of any rejected point are also rejected.

DESCRIPTION

This task sets the values of the default aperture parameters for the tasks **apedit** and **apfind** which define new apertures. For a description of the components of an aperture see the paper **The APEXTRACT Package**. In **apedit** the default aperture limits and background parameters are only used if there are no other apertures defined. The aperture identification table is used when reordering the apertures with the 'o' key. When run the parameters are displayed and modified using the **eparam** task.

The aperture limits and background fitting sample regions are defined relative to the center of the aperture. The background fitting parameters are those used by the ICFIT package. They may be modified interactively with the 'b' key in the task **apedit**. For more on background fitting and subtracting see **apbackground**.

EXAMPLES

To review and modify the default aperture parameters:

```
cl> apdefault
```

REVISIONS

APDEFAULTV2.11

The aperture ID table information may now be contained in the image header under the keywords SLFIBnnn.

SEE ALSO apbackground, apedit, apfind, icfit